

PHASE 1

RESEARCHING

Project Overview

The Gasoline Price Prediction System is a tool that helps forecast fuel prices in the Philippines one month ahead. It gathers information from different sources such as ship traffic, international oil and gas prices, and local economic growth. This data is stored in a database and processed using machine learning models to predict if prices will rise, fall, or remain stable.

The system was created to support businesses and drivers who struggle with sudden fuel price changes. By giving early forecasts, the project allows transport companies and logistics managers to plan their budgets, set delivery fees, and manage expenses more effectively.

Problem Statement

Fuel prices in the Philippines are unstable because of global market changes and local supply issues. Businesses and drivers often face problems when prices suddenly increase, making it hard to plan monthly expenses. Current tools only show real-time prices but do not give forecasts for the coming weeks.

This system solves the problem by providing a one-month forecast. It helps companies prepare for possible price changes, manage contracts better, and control operating costs. By knowing the trend early, businesses can make smarter financial decisions and avoid unexpected losses.

Target Audience

The primary target users of the proposed system are logistics managers and transport companies in the Philippines who rely heavily on fuel for their daily operations. These users need a reliable way to forecast gasoline prices one month ahead so they can plan delivery fees, manage budgets, and avoid sudden increases in operating costs. The system helps them prepare for future expenses and make smarter financial decisions.

The secondary target users are business owners, financial planners, and government analysts. Business owners and planners require accurate forecasts to negotiate contracts, stabilize monthly expenses, and reduce risks from fuel price changes. Government analysts benefit from the system by using its data to study fuel trends and support policy planning. The proposed system aims to improve financial stability and provide clear insights for both private companies and public institutions.

Competitor Analysis

Features	Accessibility	Readability	Clear Navigation	Visual Design	Functionality	Overall User Experience
International Market Trackers	✓				✓	
Economic Dashboards	✓	✓				
Real-Time Fuel Price Apps	✓	✓	✓	✓		
Gasoline Price Prediction System	✓	✓	✓	✓	✓	✓

Table 1: Comparative Matrix

A competitor analysis was conducted to evaluate existing systems and identify design practices that can improve the Gasoline Price Prediction System.

One of the competitors studied was International Market Trackers, which provide daily crude oil and natural gas prices. These platforms are functional and accessible for monitoring global energy trends. Their strength lies in giving updated international market data. However, they do not offer localized forecasts for the Philippines, which makes them less useful for businesses that need medium-term planning. The proposed system can adopt their global monitoring strengths while adding localized predictive analytics that directly benefit Philippine users.

Another competitor is Economic Dashboards, which present GDP growth and consumption data with good readability. These dashboards help users understand demand trends and economic conditions. Their weakness is that they do not integrate supply chain signals such as ship traffic or port congestion. As a result, they cannot fully capture the factors that influence gasoline prices. The proposed system can incorporate their clear data presentation while improving functionality by combining economic indicators with maritime traffic data.

Lastly, Real-Time Fuel Price Apps provide current gasoline prices at different stations. They are accessible, easy to navigate, and visually appealing. Their strength is convenience for drivers who want to know the latest pump prices. However, they only show present information and do not provide forecasts. The Gasoline Price Prediction System can adopt their user-friendly design while adding predictive features that give businesses and drivers a one-month outlook.

The Gasoline Price Prediction System stands out because it combines the strengths of all competitors' global B market data, economic indicators, and real-time updates—while adding machine learning forecasting.

This unique approach ensures businesses in the Philippines receive actionable insights for the next 30 days, improving financial planning and stability.

User Personas

[Insert Persona Card]

Figure 1: **Persona One - Maria Lopez**

Maria Lopez is a 29-year-old logistics coordinator working for a delivery company in Manila. She manages transport schedules and monitors fuel expenses for daily operations. Her main goal is to plan delivery costs ahead of time and maintain stable operating expenses. She needs a reliable system that provides accurate one-month fuel price forecasts and easy-to-read charts to help her make quick decisions. Maria often feels frustrated by sudden fuel price changes and the lack of forecasting tools for local markets. She is highly proficient in technology, using computers and smartphones daily for work, and prefers accessing the system through her laptop and mobile phone.

[Insert Persona Card]

Figure 2: **Persona Two - David Cruz**

David Cruz is a 40-year-old business owner who operates a small retail company with a fleet of delivery trucks. He closely tracks fuel expenses because they directly affect his profit margins. His goal is to forecast fuel prices accurately and plan monthly budgets for his business. He needs a dashboard that combines past data with one-month forecasts to guide his financial planning. David's main frustration is the unpredictability of fuel price hikes and the limited access to reliable forecasting tools. He has moderate technical proficiency, being comfortable with mobile apps and web dashboards, and prefers using his smartphone and tablet to manage his business operations.